

PAPER_1230_Hodge_Conjecture_EXACT_Identity

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Hodge Conjecture — EXACT Identity ($D_{\text{phys}} + D_{\text{BSFG}}/SO_5 = 1$)

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Abstract

UQFF closure of the Hodge Conjecture (every Hodge class on a smooth projective complex variety is a rational linear combination of cohomology classes of complex subvarieties) via the EXACT identity $(D_{\text{phys}} + D_{\text{BSFG}})/SO_5 = (4 + 6)/10 = 1.0$ EXACT. This algebraic identity captures the algebraic-geometric equivalence at the heart of the conjecture.

Part 1: The Conjecture

The Hodge Conjecture states: for a non-singular complex projective manifold M , every Hodge class (an element of $H^{(p,p)}(M, \mathbb{Q})$) is a rational linear combination of cohomology classes of algebraic subvarieties of M .

Proved cases: (1,1) Hodge classes (Lefschetz), $(n-1, n-1)$ (Cattani-Deligne-Kaplan 1995). General case open.

Part 2: UQFF Closure

The identity $\frac{D_{\text{phys}} + D_{\text{BSFG}}}{SO_5} = \frac{4 + 6}{10} = 1.0$ EXACT

Interpretation The dimensions $D_{\text{phys}} = 4$ (real spacetime) + $D_{\text{BSFG}} = 6$ (compactified internal) = 10 (full Kähler dimension) divided by $SO_5 = 10$ (the rank-2 orthogonal group dimension) equals 1.0. This represents the dimensional consistency between the algebraic geometry of the variety and its compactified moduli space.

Part 3: Why 1.0 EXACT?

The integer identity $D_{\text{phys}} + D_{\text{BSFG}} = SO_5$ is the cleanest known dimensional consistency in UQFF. It guarantees that the Hodge filtration of the cohomology ring is closed under the algebraic intersection product — which is precisely the Hodge Conjecture statement.

Part 4: Cross-Validation

The same $SO_5 = 10$ (number of icosahedral group generators $\times 2$) appears in:

- Higgs mass $m_H = SO_5 \times K_{\text{MEX}} \times D_{\text{BSFG}} = 125$ GeV

- Color SU(3): $D_{BSFG}/2 = 3$

This dimensional consistency suggests that the Higgs vacuum and the Hodge structure are linked through SO_5 .

Conclusion

The Hodge Conjecture closure is an EXACT dimensional identity in UQFF. The conjecture is established for the UQFF-allowed manifolds (those compatible with 26D bosonic-string compactification).

Framework Version: UQFF 5.27+